

Lab 02

Object Oriented Programming

BS DS/CY Fall 2025 Morning/Afternoon

Instructions:

- Indent your code properly.
- Use meaningful variable names and follow standard naming conventions.
- Use clear and meaningful prompts/labels for all program input and output.
- Do not discuss your problems with fellow students. If you face any difficulty in understanding, consult your TAs.

Task 01

Write a program in which you should:

- 1- Declare and initialize an integer **j** and display its address and data on screen.
- 2- Declare a pointer to integer **p** and assign the address of **j** to **p**. Then display the address and contents of **p**, compare the contents of **p** with the address of **j**.
- 3- Increment the value of **j** by using dereference operator with the pointer **p**.
- 4- Repeat the above three steps for **char** and **double** data types.

Task 02

Write a program in which you should:

- 1- Declare and initialize an integer array **arr[5]** and display the address of its first index.
- 2- Declare a pointer to integer **p** and assign the contents of **arr** to **p**. Take input from user for all of the elements of the array by using pointer **p** and dereference operator.
- 3- Display the address and contents of all the elements of the array by using pointer **p** and dereference operator.
- 4- Repeat the above three steps for **char** and **double** data types.

Task 03

Write a program that declares a **struct** Employee. Employee contains following data members:

```
char fName[20] // contains first name of the employee
char lName[20] // contains last name of the employee
int bYear // contains year of birth of the employee. Value range: 1950 to 1990
char bMonth[4] // contains month of birth of the employee e.g. JAN, JUL, DEC
int bDay // contains date (day) birth of the employee. Value range: 131
int jYear // contains joining year of employee. Value range: 1980 to 2013
char jMonth[4] // contains joining month of employee e.g. JAN, FEB, JUN, DEC
int jDay // contains joining date (day) of employee. Value range: 131
long salary // contains salary of the employee. Value Range 10000100000
```

1. Define following functions:
 - a. **initEmployee(.....)** //that initializes an object of Employee. (No user Defined Values here)
 - b. **viewEmployee(.....)** //that displays all the data members of an object of Employee.\
 - c. **addEmployee(.....)** //that takes values from user and stores it in an object of Employee.
2. Declare an object globally (Employee emp before main)
3. Write main() function contains following instructions:

```
void main()
{
    initEmployee();
    viewEmployee();
    addEmployee();
    viewEmployee();
}
```

Task 04

Modify the **Task3** in a way that now it contains following **structs**:

DateOfBirth

```
int year; // contains year of birth of the employee. Value range: 1950 to 1990
char month[4] // contains month of birth of the employee e.g. JAN, JUL, DEC
int day // contains date (day) birth of the employee. Value range: 131
```

JoiningDate

```
int year; // contains joining year of employee. Value range: 1980 to 2013
char month[4] // contains joining month of employee e.g. JAN, FEB, JUN, DEC
int day // contains joining date (day) of employee. Value range: 131
```

Employee

```
char fName[20] // contains first name of the employee
char lName[20] // contains last name of the employee
DateOfBirth dob // contains date of Birth
JoiningDate jDate // contains joining date
long salary // contains salary of the employee. Value Range 10000100000
```

1. Define following functions:
 - a. **initEmployee(.....)** that initializes an object of Employee. (No user Defined Values here)
 - b. **viewEmployee(.....)** that displays all the data members of an object of Employee.
 - c. **addEmployee(.....)** that takes values from user and stores it in an object of Employee.
2. Declare an object globally (Employee emp before main)
3. Write main() function that contains following instructions:

```
void main()
{
    initEmployee();
    viewEmployee();
    addEmployee();
    viewEmployee();
}
```

Task 05

Extend the **Task4** in a way that now it contains:

1. Array of employees i.e.

```
const int noOfEmployee = .....;
Employee emp [ noOfEmployee ];
```
2. A function **searchEmployee(....)** that takes last name of an employee, and finds it in the array. If found then displays all of its information.